

Abstract

CiviVision is a technology-driven civic issue management platform designed to bridge the communication gap between citizens and local authorities. Public infrastructure concerns like potholes, waste dumps, broken streetlights, and drainage issues are often difficult to report, track, and resolve efficiently. CiviVision addresses this challenge by providing a centralized full-stack digital platform through which citizens can report grievances and monitor progress. The proposed system provides a structured mechanism enabling complaints to be classified, prioritized, and tracked. Built as a decoupled web application using React (Vite) on the frontend and Express/Prisma on the backend, the platform supports geotagging, image-based validation, and duplication checking to improve data integrity. It features real-time geospatial heatmaps for administration and one-click SOS triggers for immediate public safety dispatches. By transforming civic complaints into structured, trackable data, CiviVision promotes smart governance, transparency, and efficient municipal service delivery.